PHYS 4041/5041:

Mathematical Methods of Physics 1

Partial Differential Equations

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Partial Differential Equations

- Differential equations with multiple independent variables. See Boas, chapter 13.
- Physical variables: x, y, z, t.
- Cartesian coordinates:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \, .$$

- Spherical coordinates:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \, ,$$

note

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\Psi}{\partial r}\right) = \frac{1}{r}\frac{\partial^2}{\partial r^2}\left(r\Psi\right) \, .$$

Named Equations

- $\nabla^2 \Psi = 0$ Laplace equation, homogeneous.
- $\nabla^2 \Psi = f(\vec{\boldsymbol{r}}, t)$ Poisson equation, inhomogeneous.
- $\nabla^2 \Psi = a \frac{\partial \Psi}{\partial t}$ diffusion equation.
- $\left[-\frac{\hbar^2}{2\mu} \nabla^2 + V(\vec{\boldsymbol{r}}) \right] \Psi = E \Psi$ Schrödinger equation.
- $\left[-\frac{\hbar^2}{2\mu} \nabla^2 + V(\vec{\boldsymbol{r}}, t) \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t}$ Schrödinger equation.
- $(\nabla^2 + k^2) \Psi = 0$ Helmholtz equation.
- $\nabla^2 \Psi = \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2}$ wave equation.

Two types of problems

- Boundary conditions given, need solution. Example: electrostatic potential.

- Eigenvalue problem. Boundary conditions given, but physically-reasonable solutions exist only for particular parameter values. Example: hydrogen atom.

- The choice of coordinate system (including the origin and orientation) will depend on the the symmetries of the problem (including the boundary conditions).

Separation of variables

- We will focus on this technique.
- Assume that equation will *separate* in the chosen coordinate system. Example: in Cartesian coordinate, try $\Psi = X(x)Y(y)Z(z)$.
- See if the equation *separates* – this will be discussed in a moment.
- Apply boundary conditions to the separated pieces of the equation.
- Build the complete solution using a linear combination of separated solutions.

Separation of variables, Cartesian coordinates

- Cartesian: $\frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} + \frac{\partial^2\Psi}{\partial z^2} + k^2\Psi = 0$ (Helmholtz equation).
- try $\Psi = X(x)Y(y)Z(z)$:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

- Key idea: x, y, z are arbitrary $\rightarrow$ each term must be constant!

$$\text{e.g.} \quad \frac{1}{X}\frac{d^2X}{dx^2} = \pm\alpha^2 \quad \frac{1}{Y}\frac{d^2Y}{dy^2} = \pm\beta^2 \quad \frac{1}{Z}\frac{d^2Z}{dz^2} = \pm\gamma^2$$

X, Y, and Z must be either sinusoidal or exponential functions of x, y, and z, respectively. We also must have $k^2 = \mp\alpha^2 \mp \beta^2 \mp \gamma^2$, in order to solve the equation.

Separation of variables, Cartesian coordinates

- Assume that we have a cube of side L, such that $0 \leq x \leq L$, $0 \leq y \leq L$, and $0 \leq z \leq L$. Further assume that Ψ vanishes on all surfaces of the cube. Find the solution to $(\nabla^2 + k^2)\Psi = 0$ (Helmholtz equation) for this situation. Note that this could also be the description of a particle in a cube via the Schrödinger equation.
- We must have sinusoidal solutions, more specifically

$$X \propto \sin \alpha x \qquad Y \propto \sin \beta y \qquad Z \propto \sin \gamma z.$$

- The boundary imply $\alpha_\ell = \frac{\ell\pi}{L}$, $\beta_m = \frac{m\pi}{L}$, and $\gamma_n = \frac{n\pi}{L}$, with ℓ, m, and n limited to $1, 2, 3, \ldots$.
- This is thus an eigenvalue problem, with $k^2_{\ell mn} = \frac{\pi^2}{L^2}(\ell^2 + m^2 + n^2)$, and $\Psi_{\ell mn} = \left(\frac{2}{L}\right)^{3/2} \sin \frac{\ell\pi x}{L} \sin \frac{m\pi y}{L} \sin \frac{n\pi z}{L}$.

Remarks on separation of variables

- The Laplacian is separable in our thee standard coordinate systems: Cartesian, cylindrical, and spherical. It is also separable in some others, such as parabolic cylindrical coordinates. However, it is unlikely to separate an arbitrary coordinate system that you might dream up.
- The choice of the sign of the separation constant(s) is important. Normally, a negative sign implies an oscillatory solution, while a positive sign implies an exponentially increasing or decreasing solution.
- There are other important issues with PDEs, that we have not addressed:
  - Uniqueness of solutions?
  - When is the problem under-determined or over-determined?

Vector calculus: Divergence Theorem

- The divergence theorem of vector calculus is the three-dimensional analog of $\int_a^b \frac{df}{dx}\,dx = f(b) - f(a)$:

$$\int_V \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{A}}\, d^3x = \oint_S \vec{\boldsymbol{A}} \cdot \hat{\boldsymbol{n}}\, dA,$$

  where the surface S bounds a volume V, and $\hat{\boldsymbol{n}}$ points out of the volume V.

- Using $\vec{\boldsymbol{\nabla}} \cdot (\phi \vec{\boldsymbol{\nabla}} \Psi) = \phi \nabla^2 \Psi + \vec{\boldsymbol{\nabla}} \phi \cdot \vec{\boldsymbol{\nabla}} \Psi$, one finds Green's first identity, which a three-dimensional analog of integration by parts:

$$\int_V (\phi \nabla^2 \Psi + \vec{\boldsymbol{\nabla}} \phi \cdot \vec{\boldsymbol{\nabla}} \Psi)\, d^3x = \oint_S \phi \vec{\boldsymbol{\nabla}} \Psi \cdot \hat{\boldsymbol{n}}\, dA.$$

Uniqueness of solutions to the Poisson equation

- Suppose that Ψ_1 and Ψ_2 are both solutions to the Poisson equation $\nabla^2 \Psi = f(\vec{\boldsymbol{x}})$ in a volume V and that both solutions are equal of the boundary surface S.
- Consider Green's first identity with $\phi = \Psi = \Psi_1 - \Psi_2$. One has $\nabla^2 \phi = 0$ in V and $\vec{\boldsymbol{\nabla}} \phi = 0$ on S. We thus have

$$\int_V |\vec{\boldsymbol{\nabla}} \phi|^2 \, d^3x = 0.$$

- The difference between the solutions is at most a constant. But considering the boundary, the constant must be zero.
- We thus conclude that the solution to the Poisson (and Laplace) equation in a volume that satisfies given values on the bounding surface(s) is unique.